



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Centre for Professional and
Continuing Education

(SCTP) Advanced
Professional Certificate
Data Science and AI



3.2 Probability and Statistics

Module Overview

3.1 Introduction to Machine Learning

3.2 Probability and Statistics for Machine Learning

3.3 Supervised Learning

3.4 Supervised Learning - Advanced

3.5 Unsupervised Learning

3.6 Time Series Data & Forecasting

3.7 Neural Network & Deep Learning

3.8 Computer Vision (CV)

3.9 Natural Language Processing (NLP)

3.10 NLP - Advanced

Module 3 · Machine Learning & GenAI

L02

How Sure *Are We?*

Probability & Statistics for ML · Sarah's week of defending every number

Where We Left Off

Priya's question is the engine of this lesson

Priya, at the end of 3.1:

"How sure are we? The model put a label on every review.

How do we know it isn't confidently wrong in ways we can't see?"

Three numbers Sarah needs to defend by Friday:

60%

of reviews are positive

→ **Distribution analysis (Tuesday)**

84%

model accuracy

→ **Confidence interval (Wednesday)**

?

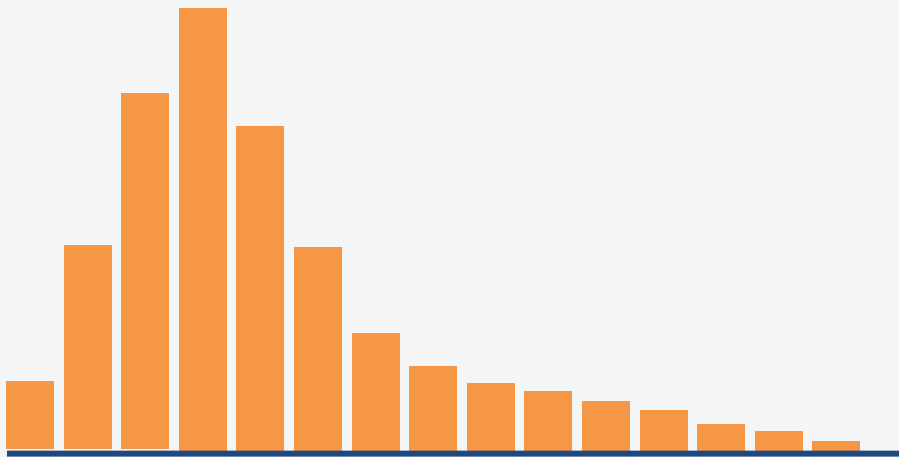
coupon reduces churn

→ **A/B test + p-value (Thursday)**

Sarah's Monday — The Polarity Histogram

Reading the shape of 10,000 reviews

What Sarah sees:



Most scores cluster in the mildly-positive range, with a long tail of very positive reviews on the right.

Sarah's first reaction:

"It's not a bell curve. There's a hump on the left, a long tail to the right."

"How do I describe this to Priya in a way that's honest?"

The vocabulary she needs:

- Skewness
- Mean vs Median
- Z-scores

Why This Matters For Sarah's Report

Shape determines which summary number to quote

Sarah's choice point

"Do I report the MEAN positivity (pulled up by the long tail) — or the MEDIAN (more representative of a typical review)?"

Report the MEAN

Sensitive to extremes. The long tail pulls it up.

✗ **Overstates how positive 'typical' reviews are**

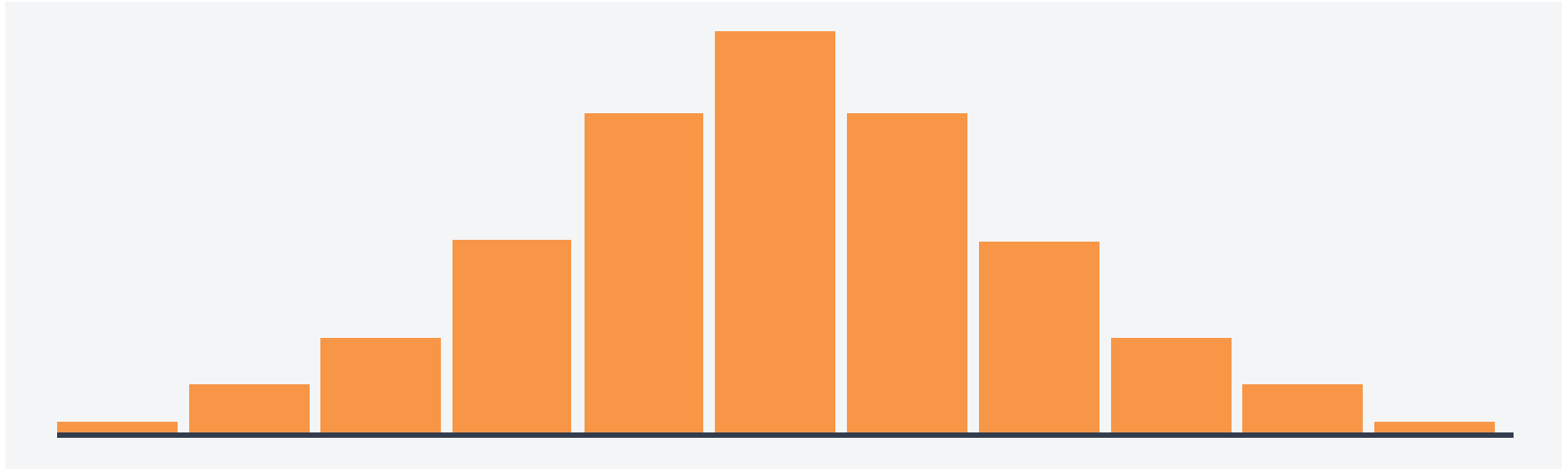
Report the MEDIAN

Robust to extremes. The 50% mark.

✓ **Better representation of a 'typical' review for skewed data**

Distribution Shape 1: **Normal (Bell Curve)**

Symmetric · Peak in the middle · Thin tails



Real-world examples

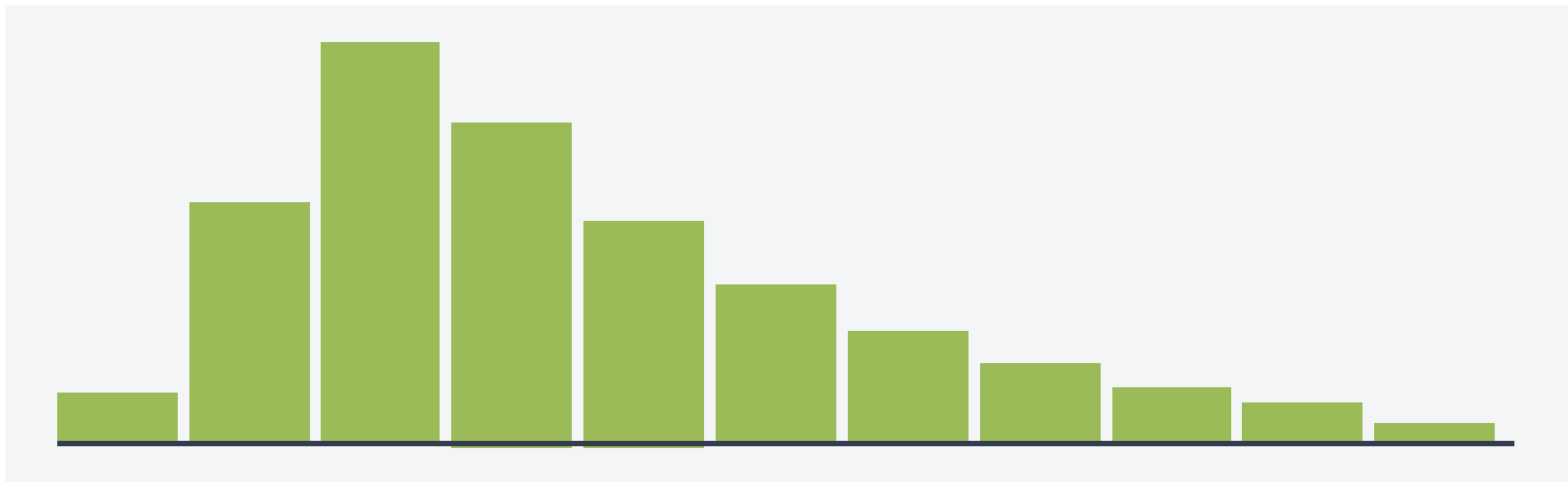
Heights of adults · Test scores ·
Measurement errors

Rule of thumb

Mean $\approx$ Median

Distribution Shape 2: **Right-Skewed**

Peak on the LEFT · Long tail to the RIGHT



Real-world Examples

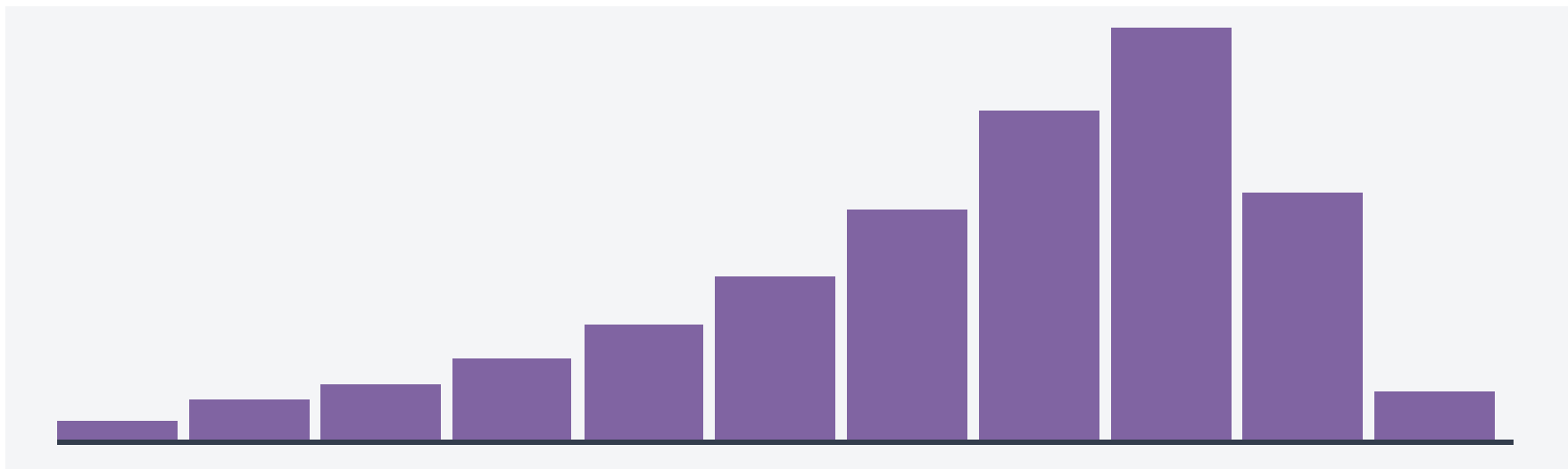
Incomes · House prices · Review lengths ·
Hospital stays

Rule of thumb

Mean > Median (mean is pulled up by the tail)

Distribution Shape 3: **Left-Skewed**

Peak on the RIGHT · Long tail to the LEFT



Real-world examples

Easy exam scores · Age at retirement ·
Lifetime of products

Rule of thumb

$\text{Mean} < \text{Median}$ (mean is pulled down by the tail)

Mean vs Median

The gap tells you about the skew

Worked example — UK household income (right-skewed)

£35k

Median

£52k

Mean

Three quick rules

Mean = Median → Data is symmetric. Quote either.

Mean > Median → Right-skewed. Quote MEDIAN for 'typical'; quote MEAN for 'total income generated'.

Mean < Median → Left-skewed. Same logic, flipped.

Back to Sarah's Reviews

Why the choice matters for her Friday deck

If Sarah quotes only the mean...

*"Average polarity is +0.12 — **mostly positive**."*

→ Priya thinks customers are happy. Hides the long left tail of complaints.

If Sarah quotes the median + shows the distribution...

*"Median polarity is +0.06 — **slightly positive**."*

*The distribution has a long tail of **very positive** reviews,
but about 40% sit near 0 (**lukewarm**) and several hundred are **strongly negative**.
Those are the ones we should look at."*

→ Priya understands the actual shape. Sarah's judgement is credible.

Z-scores — Putting Values in Context

How many standard deviations from average?

Formula

$$z = (\text{value} - \text{mean}) / \text{standard deviation}$$

Worked example — Sarah's exam score

Sarah scored 85 on a test. Is that good?

If class average = 60, std = 10 $z = (85-60)/10 = 2.5$ → Top ~1% of the class

If class average = 82, std = 2 $z = (85-82)/2 = 1.5$ → Above average, not exceptional

Same raw score (85) means very different things in different contexts. Z-scores reveal the difference.

Two Ways Z-scores Show Up in ML

Outlier detection · Feature standardisation

1. Outlier detection

Rule of thumb: $|Z| > 3$ = investigate.

For Sarah's reviews:

$Z < -3$ → an unusually NEGATIVE review.
Route to customer service.

$Z > +3$ → an unusually POSITIVE review.
Maybe a sycophant or a bot.

Works on any quantitative variable — model accuracy across stores, transaction amounts, log latencies.

2. Feature standardisation

Before training a model, transform every numeric feature: age, income, days-since-last-login, ...

Each becomes **mean=0, std=1**.

Why? Many algorithms (linear regression, k-NN, neural nets) assume features are on similar scales.

In sklearn:

```
StandardScaler().fit_transform(X)
```

We'll use this in L03 onwards.

● CODE-ALONG

Open: `02_distributions.ipynb`

What you'll do

1. Plot Sarah's polarity & review-length distributions side-by-side.
2. Identify the shape and compute mean vs median for each.
3. Spot which summary statistic to quote for which question.

🕒 30 minutes · Stop at the ● Extension marker · Extension is self-paced after class

Sarah's Wednesday — 84% From 200 Reviews

Is it 'exactly' 84%? Or is that just one sample's answer?

Sarah's plan to validate the L01 sentiment model:

Hand-label 200 reviews herself. Compare to the model's predictions. Report agreement rate.

The result

167 of 200 reviews agreed.

That's 83.5% — rounded to:

84%

Priya's follow-up

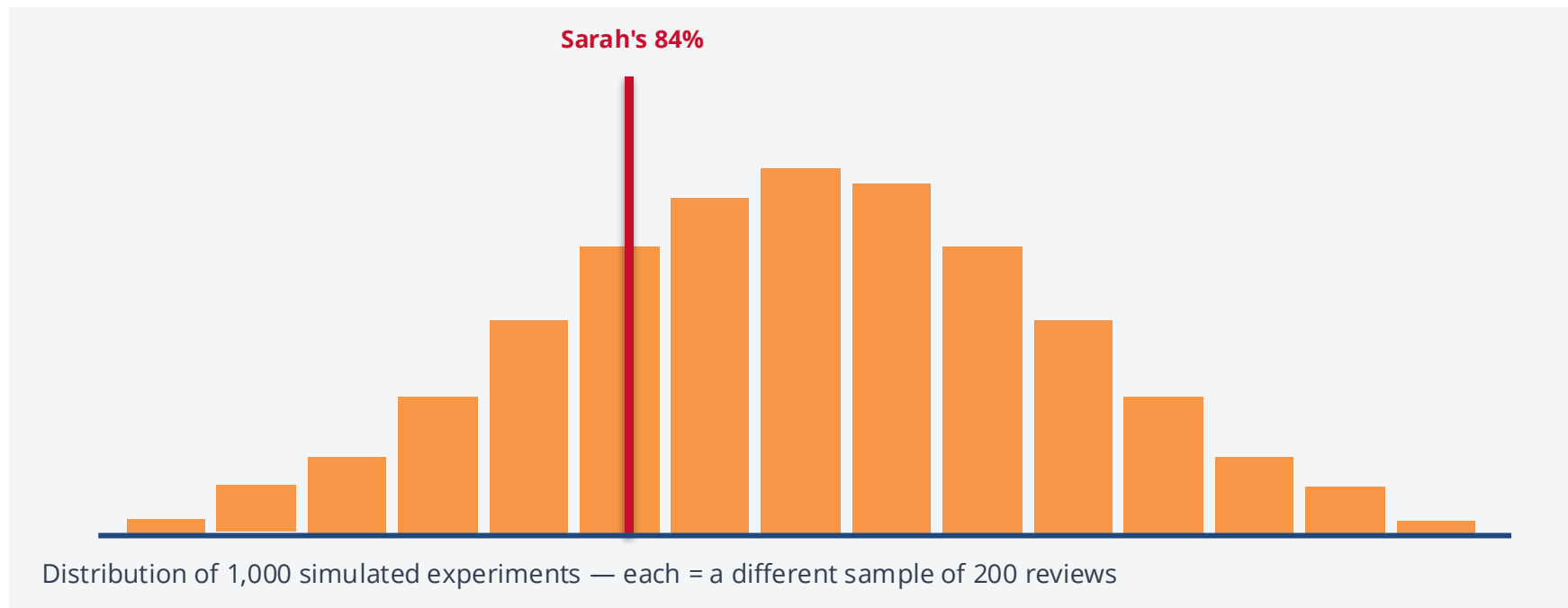
"So the model is EXACTLY 84% accurate?"

Sarah pauses. If she had labelled a different 200 reviews, would she get the same number?

Probably not. But how close?

Simulating Sarah's Sample

What if she'd labelled a DIFFERENT 200?



Bell-shaped. Most estimates fall between **78%** and **89%**. Sarah's **84%** is one draw from this distribution.

The Central Limit Theorem (CLT)

Why Sarah's histogram looks bell-shaped

The Central Limit Theorem, in plain English

*"Even if the underlying data isn't normally distributed, the **AVERAGE** of many samples from that data will follow a normal distribution. The larger each sample, the tighter the normal becomes."*

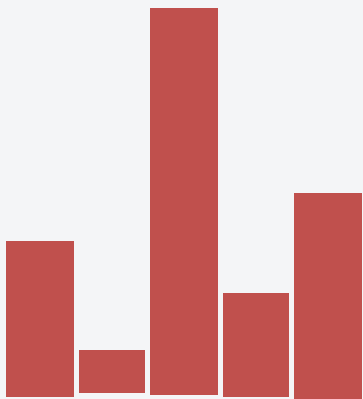
Three things this lets you do

- 1 Compute confidence intervals on almost any accuracy metric**
(model accuracy, conversion rate, click-through rate, complaint rate)
- 2 Run hypothesis tests without assuming the raw data is normal**
(t-test, z-test, A/B tests — all rely on the CLT)
- 3 Reason about sample size**
(want a tighter estimate? Increase n . The bell curve gets narrower.)

The Skittles Analogy

The raw data isn't normal — the AVERAGES are

1. The jar



Not normal — colours unequal

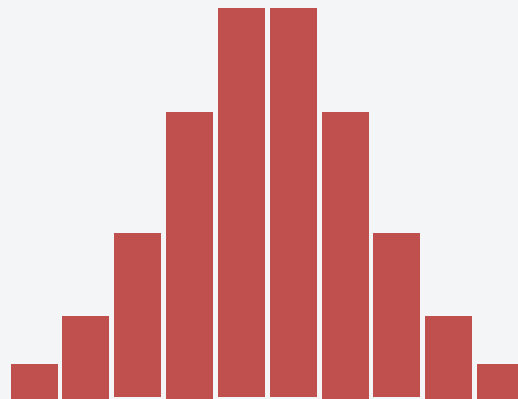
2. Take handfuls of 30

Each handful gives a different red-count.

12, 9, 14, 11, 13, 10, 15,
11, 12, 13...

(Do this 1,000 times.)

3. Plot the averages



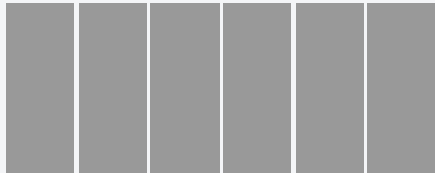
Bell-shaped. Always.

Same logic for Sarah: each 'sample of 200 reviews' is like a handful of Skittles.
The accuracy rates follow a bell curve.

CLT — Visual Proof in Three Panels

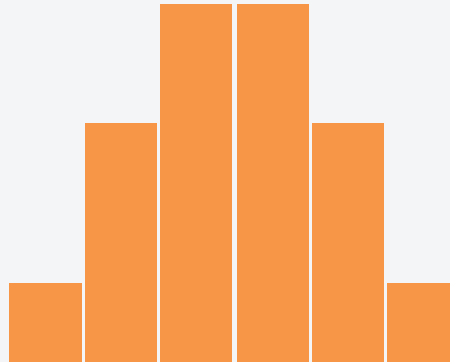
One die · five dice · thirty dice

$n = 1$ die



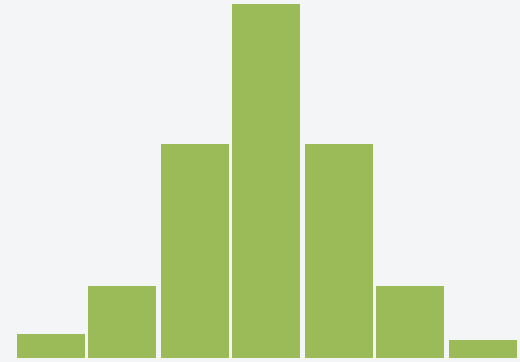
Flat. Not normal.

$n = 5$ dice (averaged)



Starting to hump.

$n = 30$ dice (averaged)



Bell curve. CLT.

As n grows, the distribution of averages narrows around the true mean — and the shape becomes **bell-shaped**.

Confidence Intervals

The bracket around every accuracy number

Plain English

*"A 95% confidence interval is a range that captures the true value about 95% of the time, **IF** we were to repeat the experiment many, many times."*

Formula for a proportion (like accuracy)

$$\text{Lower} = \hat{p} - 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad \text{Upper} = \hat{p} + 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Sarah's 84% from 200 reviews

$$\hat{p} = 0.84, \quad n = 200 \rightarrow \text{CI} = [78.4\%, 89.6\%]$$

The CI Mis-reading

The phrasing trap everyone falls into

✗ Wrong (the everyday mis-reading)

"There is a 95% probability that the true accuracy is between 78% and 89%."

Why wrong: the true accuracy is a single fixed number. It doesn't have a probability of being in a range.

✓ Right (the careful phrasing)

"If we were to repeat this 200-review labelling exercise many times, about 95% of the confidence intervals we'd compute would contain the true accuracy."

What has the 95% property: the METHOD. Not this specific interval.

In practice, you can say: *"We're 95% confident the accuracy is between 78% and 89%"* — as a shorthand. Just know what it really means.

● CODE-ALONG

Open: `03_confidence_intervals.ipynb`

What you'll do

1. Run Sarah's 200-review experiment 1,000 times — watch the CLT emerge.
2. Compute the 95% CI for her 84% accuracy from first principles.
3. See how sample size affects the width — the 'square root rule'.

🕒 30 minutes · Stop at the ● Extension marker · Extension is self-paced after class

Thursday — Aisha's Apology Coupon Idea

Will it reduce complaints — or is it expensive noise?

Aisha · Customer Service

Send a 10% apology coupon to every NEGATIVE-tagged customer.
She believes it'll turn some unhappy customers around.

Marcus · CTO

Thousands of pounds a week + engineering work to wire the trigger.
He wants evidence before signing off.

Sarah · Data Analyst

Test it properly. Define the hypothesis. Run an A/B test. Compute a p-value.
Her L02 toolkit makes her the right person to answer.

Framing the Test — Before Any Code

Four things to write down

Null hypothesis (H_0)

Coupon has NO effect on 30-day complaint rate.

Alternative hypothesis (H_1)

Coupon REDUCES 30-day complaint rate.

Outcome metric

30-day complaint rate (leading indicator of churn — available in 30 days vs 90+).

Two groups

Control: NEGATIVE-tagged customers, no coupon.
Treatment: same, but receive the coupon.

Why 30-day complaint rate, not churn?

Complaint data lands in 30 days; churn requires 90+ days of observation.

The P-value

The probability that the data is just noise

Plain English

" $P = P(\text{data this extreme} \mid \text{null hypothesis is true})$. If p is small, the data would be unlikely under H_0 , so we reject H_0 ."

Standard decision rule ($\alpha = 0.05$):

$p < 0.05$

Reject the null hypothesis.

The observed difference is unlikely under H_0 .
There's evidence of a real effect.

$p \geq 0.05$

FAIL to reject the null hypothesis.

Not enough evidence to declare an effect.
(NOT the same as 'no effect' — see next slide.)

The P-value

The p-value answers a crucial question: "If there really was no effect, what's the probability of seeing results this extreme or more extreme by pure chance?"



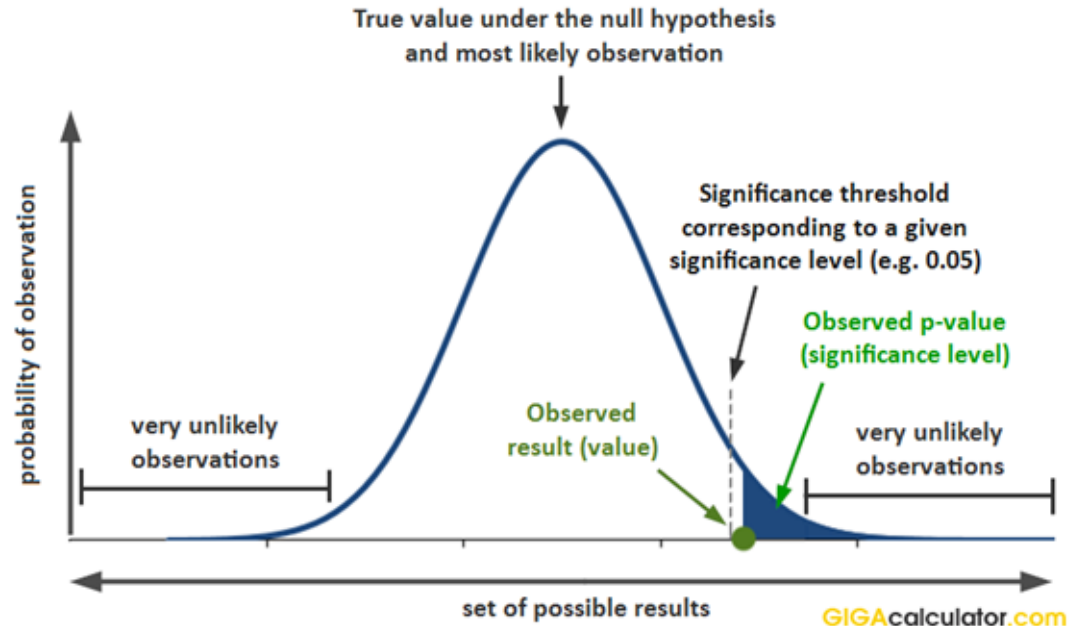
$P < 0.05$

Statistically significant - less than 5% chance results are due to random luck



$P > 0.05$

Not statistically significant - could easily be due to chance



<https://www.gigacalculator.com/articles/p-value-definition-and-interpretation-in-statistics/>

P-value Mis-readings — Part 1 of 2

Two common ways to misquote a p-value

✗ Mis-reading 1

' $p = 0.03$ means there is a 3% chance the null hypothesis is true.'

What p actually means: $P(\text{data this extreme} \mid H_0 \text{ is true})$. Not $P(H_0 \mid \text{data})$.

The flipped form is Bayesian and requires a different framework. The everyday p-value is NOT a probability over hypotheses.

✗ Mis-reading 2

' $p < 0.05$ means the effect is large and important.'

Statistical significance $\neq$ practical (or economic) significance.

With a large enough sample, even a tiny meaningless difference will produce $p < 0.05$. Always report effect size alongside the p-value.

P-value Mis-readings — Part 2 of 2

The mis-reading that kills good ideas

✗ Mis-reading 3

'p = 0.08, not significant — so the coupon doesn't work.'

Why it's wrong:

'Failed to reject H_0 ' $\neq$ ' H_0 is true'. The effect might be real but the sample was too small to detect.

Absence of evidence is NOT evidence of absence.

The careful phrasing

'We did not find statistically significant evidence of an effect at this sample size. We cannot conclude the coupon has no effect. We recommend extending the test or increasing the sample size to reduce uncertainty.'

● CODE-ALONG

Open: **04_ab_testing.ipynb**

What you'll do

1. Frame the test, then simulate the 2-week experiment.
2. Compute the two-proportion z-test from first principles $\rightarrow$ p-value.
3. Practice the three corrected readings on the actual numbers.

🕒 20 minutes · Stop at the ● Extension marker · Extension is self-paced after class

Friday — The Three Defended Numbers

Same numbers, now backed by the right tools

60%

of reviews are positive

Distribution analysis ·
Median + chart of skew

84%

model accuracy

Confidence interval ·
95% CI: [78%, 89%] from n=200

~6pp

drop in 30-day complaint rate from
the coupon ($p < 0.05$)

A/B test · Two-proportion z-test

Priya: *'Looks great. But you used a pre-trained model. Can you build one TRAINED on NorthStar's own data?'*

Bridge to L03

Marcus, after the Friday meeting:

"OK. The sentiment model holds up. But you used a pre-trained one — off the shelf.

Can you build us a model that predicts CHURN from our own customer data?

Something trained on NorthStar behaviour, not some generic internet corpus?"

Lesson 3 — Supervised Learning Foundations

Train/test split · preprocessing · regression vs classification · linear & logistic models.

First time you train a model end-to-end on NorthStar's own data.

Assignment due before L03 — same workflow, new dataset (Lakeside Bank).

Q&A — and What's Next

Common questions · assignment preview



Common questions

- When should I use a t-test vs a z-test?
- How big should my A/B test sample be?
- How do I report 'no significant difference' to a stakeholder?
- Why 1.96 in the CI formula — where does that come from?
- What if the underlying data is really not normal?

Bring your own questions!



Assignment preview

Open: [notebooks/assignment.ipynb](#)

Lakeside Bank — Tom Bradley needs help.

Apply the same three tools:

- Distribution analysis
- Confidence intervals
- A/B test with p-value

— but to a new dataset and a new business problem.

See you in L03.

End of Lesson - Exit Ticket

Survey Link

Slido link will be provided in class

